

ARK-488 — Authority Engine · P95 Decision Latency

Status: PREREGISTRATION (locked before execution)

Experiment Date: 2026-07-18

Series: ExecutionProof P02 Latency/Throughput/Scale

Question

On a warm reference **Authority Engine**, what is the per-decision latency distribution (p50 / p95 / p99) for a single current-state authority check?

This is the buyer's most-asked question applied to the authority half of Verification-Before-Execution: how much latency does an at-execution "does this principal still hold this authority?" check add to the critical path?

Scope & Honesty Boundary

Component under test is the minimal in-process reference `AuthorityEngine` — built to MEASURE, not the production engine (persistent store, replication, external IdP). Claims bounded to this reference under the stated load.

Hypothesis

Warm per-decision latency **p95 ≤ 50 μs** and **p99 ≤ 200 μs** (dict + set membership operations, no I/O).

Component Under Test

Reference `AuthorityEngine` (exact, fail-closed): principal lookup → grant membership → current-state revocation test. `engine/perf_harness.py --dimension p95_latency`.

Methodology

- **Warmup:** 10,000 decisions (excluded)
- **Measurement:** 200,000 timed decisions on the valid grant path
- Record per-call elapsed time (μs); compute p50, p95, p99, mean, max

Metrics

Primary: `latency_us` — p50, p95, p99

Secondary: mean, max

Thresholds (Gate Conditions)

C1 (Latency): $p95 \leq 50 \mu s$

C2 (Correctness): correctness gate must PASS (valid→ALLOW, mutated→DENY, revoked→DENY)

Verdict: PASS if $C1 \wedge C2$; otherwise FAIL

Kill-Gate

K1: Correctness gate failure → GATE-STOP; latency numbers void.

Honest Findings Commitment

Any deviation (tail latency spikes, correctness failures) disclosed in RESULTS.md.

Execution Protocol

1. **Lock** prereg + MANIFEST.txt hashes
 2. **Execute** `engine/perf_harness.py --dimension p95_latency`
 3. **Record** to `results/p95_results.json`
 4. **Evaluate** against C1, C2
 5. **Report** in RESULTS.md
-

Preregistered: 2026-07-18

Investigator: Remnant Fieldworks Inc. / ExecutionProof Research Program

Covenant: RF Standing Covenant (preserve outcomes, bound claims, honest disclosure)